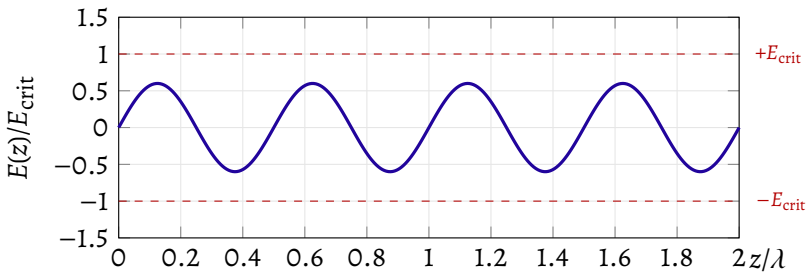


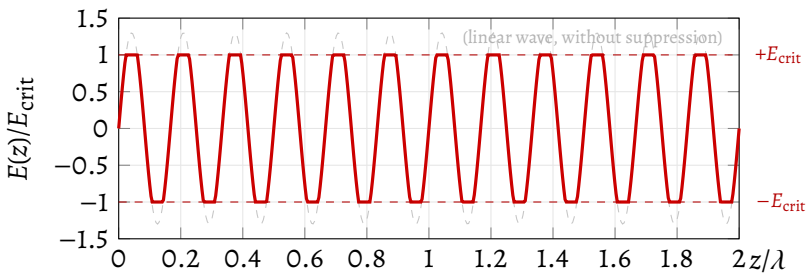
$f < f_{\text{crit}}$  — linear regime



Field never reaches  $E_{\text{crit}}$  · No semi-particle formed · No electron ejected

(a) Linear regime:  $f < f_{\text{crit}}$  (low frequency)

$f > f_{\text{crit}}$  — nonlinear regime activated



Field reaches  $E_{\text{crit}}$   $\Rightarrow$  nonlinear suppression · Semi-particles formed at the peaks · Ejection if the energy exceeds  $\phi$

(b) Nonlinear regime:  $f > f_{\text{crit}}$  (high frequency)

Figure 9 – Two regimes of *formation* of semi-particles under the hypothesis of restitutive nonlinearity (cf. Appendix R). **(a)** For  $f < f_{\text{crit}}$  (for a fixed amplitude  $V_0$ ), the amplitude remains below  $E_{\text{crit}}$  (dashed red): linear regime, with no semi-particles and hence no ejection of electrons. **(b)** For  $f > f_{\text{crit}}$ , the field would reach amplitudes above  $E_{\text{crit}}$  (gray ghost), but the nonlinear suppression limits the peaks (solid red curve), forming localized excitations (semi-particles); each one ejects an electron if its energy exceeds the work function of the metal.

